

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

QUIZ 1 SET D

SUMMER SEMESTER, 2023-2024

DURATION: 25 MINUTES

FULL MARKS: 20

CSE 4205: Digital Logic Design

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

ID: _____

1. A number $(891)_b$ in an unknown base b is equivalent to the octal number $(4642)_8$. Determine the base b .

3
(CO1)
(PO1)

Solution:**Given:**

$$(891)_b = (4642)_8$$

Step 1: Convert octal to decimal

$$(4642)_8 = 4 \cdot 8^3 + 6 \cdot 8^2 + 4 \cdot 8^1 + 2 \cdot 8^0 = 4 \cdot 512 + 6 \cdot 64 + 4 \cdot 8 + 2 = 2048 + 384 + 32 + 2 = 2466$$

Step 2: Convert $(891)_b$ to decimal and solve

$$(891)_b = 8b^2 + 9b + 1 = 2466 \Rightarrow 8b^2 + 9b - 2465 = 0$$

Using the quadratic formula:

$$b = \frac{-9 \pm \sqrt{9^2 - 4 \cdot 8 \cdot (-2465)}}{2 \cdot 8} = \frac{-9 \pm \sqrt{81 + 78880}}{16} = \frac{-9 \pm \sqrt{78961}}{16} = \frac{-9 \pm 281}{16}$$

Evaluating both roots:

$$b = \frac{-9 + 281}{16} = \frac{272}{16} = \boxed{17}$$
$$b = \frac{-9 - 281}{16} = \frac{-290}{16} = -18.125 \quad (\text{Not valid})$$

Final Answer:

$$\boxed{b = 17}$$

Rubrics:

- 1 point for correctly converting the two numbers to a known base.
- 1 point for correctly expressing the given numbers into a valid equation.
- 1 point for correctly determining the root of the equation and neglecting the invalid root.

2. Convert $(26)_{16}$ into Gray Code.

3
(CO1)
(PO1)

Solution:

At first convert the given number into binary number,

$$(26)_{16} = (100110)_2$$

Final Answer:

$$(100110)_2 = (110101)_{Gray}$$

Rubrics:

- 1 point for correctly converting the decimal number into binary number.
- 1 point for writing down relevant equation/expression to convert binary number to gray code.
- 1 point for correct gray code.

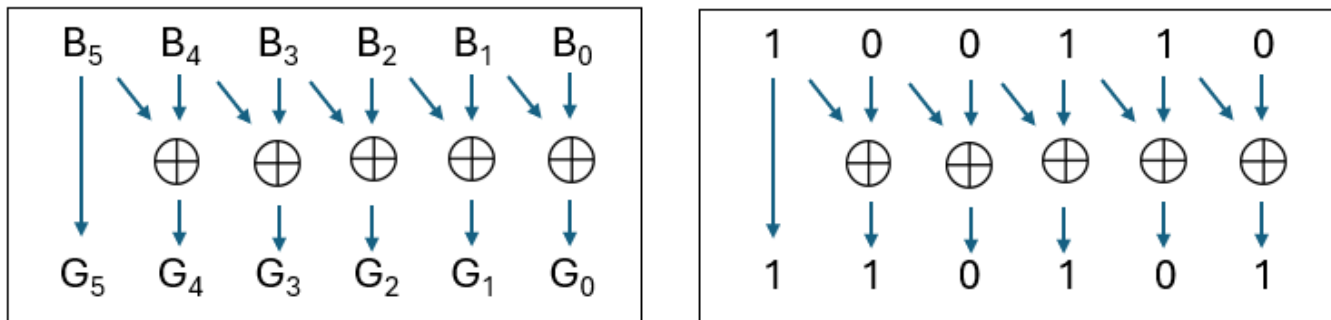


Figure 1: Binary to Gray

3. Subtract $(645)_9$ from $(867)_9$ using *Diminished Radix Complement*. (Use a 4 digit register.)

6
(CO2)
(PO1)

Solution:

Given,

$$(867)_9 - (645)_9$$

Convert to 4 digit register compatible

$$= (0867)_9 + (-0645)_9$$

Converting $(0645)_9$ to (9-1) or 8's Complement (subtract each digit from 8),

$$8\text{'s Complement of } (0645)_9 = (8243)_{8\text{'s Complement}}$$

So,

$$(867)_9 - (645)_9 = (222)_9$$

Final Answer:

$$(222)_9$$

Rubrics:

- 1 point for using 4 digit representation.
- 1 point for representing the subtraction operation in terms of addition.
- 1 point for correctly performing 8's complement.
- 2 points for correct addition using 8's complement.
- 1 point for correctly handling the overflow bit.

		1	1	1		
$(0867)_9$	→	0	8	6	7	
$(0645)_9$'s 8's complement	→	8	2	4	3	
	(+)					
		1	0	2	2	1
		(overflow)				
		DISCARD				
		Add 1				
					+	1
				2	2	2

$\therefore (867)_9 - (645)_9 = (222)_9$

Figure 2: 8's Complement

4. Simplify the following Boolean expression using a Karnaugh Map (K-map):

8
(CO3)
(PO2)

$$F(A, B, C, D) = \prod M(3, 5, 6, 9, 12) + \sum d(2, 10, 14)$$

Where:

- $\prod M(\dots)$ denotes a product of maxterms.
- $\sum d(\dots)$ denotes the don't-care conditions.

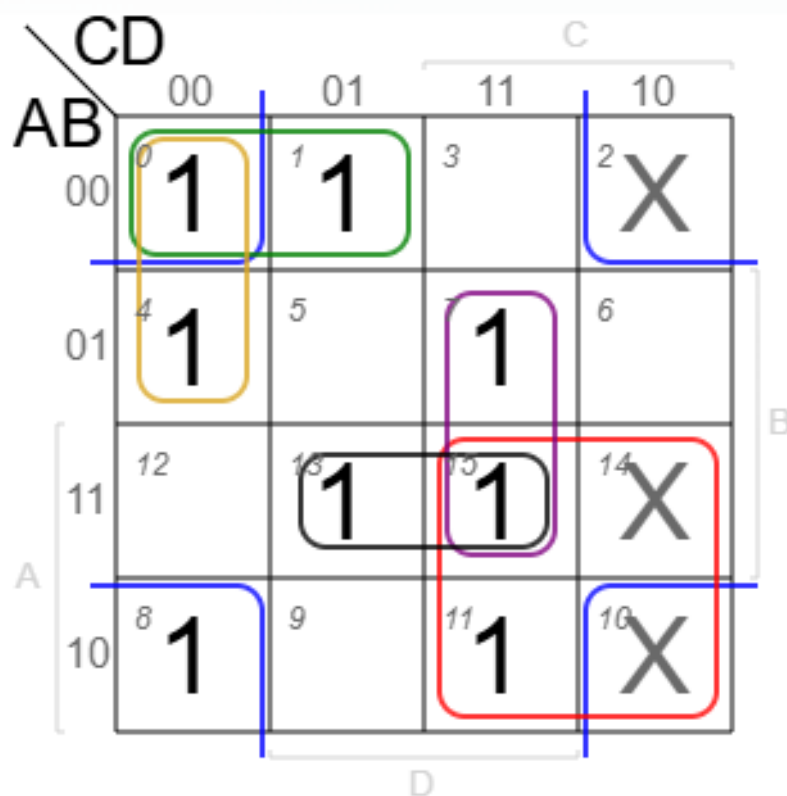
While solving, draw the K-Map and identify the groups(minterms).

Solution:**Final Answer:**

$$F(A, B, C, D) = B'D' + AC + A'B'C' + A'C'D' + BCD + ABD$$

Rubrics:

- 2 point for building the 4 variable K-map.
- 1 point for correctly identifying each essential prime component (times 6).



$$F = B'D' + AC + A'B'C' + A'C'D' + BCD + ABD$$

Figure 3: K-map